The Circuit Imprint:
Loss-Surface Geometry Fingerprints the Grokking Transition

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Abstract

Grokking—where a network generalises long after memorising its training set—involves an algorithmic switch from a dense lookup table to a sparse Discrete Fourier Transform (DFT) circuit. We dissect this transition with three complementary geometric probes: (i) the SGLD-estimated Local Learning Coefficient (LLC), sensitive to *global multiplicity*; (ii) the Hessian trace $\mathrm{tr}(H)$, sensitive to *total local curvature*; and (iii) their ratio $\hat{\lambda}_H = \mathrm{tr}(H)/2\lambda_{\max}$, which approximates *effective circuit dimensionality*. Across five random seeds, $\mathrm{tr}(H)$ collapses $\sim 4\times$ at grokking while SGLD-LLC is entirely unchanged—establishing that grokking is a local-curvature event invisible to the global probe. Post-grokking, $\hat{\lambda}_H$ settles at $\sim 20 \pm 2$, directly fingerprinting the Fourier circuit's intrinsic dimensionality (~ 11 active frequencies $\times$ 2 free parameters $\approx$ 20). A direct Fourier-amplitude analysis of the token embeddings confirms ~ 11 active frequencies independently at the same training step, demonstrating that loss-surface geometry recovers circuit complexity without any task-specific identification.

1. Introduction

Grokking (Power et al., 2022) offers a controlled setting in which a transformer trained on modular arithmetic $(a+b) \bmod p$ memorises its training set, then—after a long plateau—abruptly generalises. Mechanistic interpretability has shown that this involves an algorithmic switch: a dense lookup table is replaced by a compact DFT circuit using only a few active Fourier frequencies (Nanda et al., 2023).

Singular Learning Theory (SLT) (Watanabe, 2009) predicts that this switch leaves a geometric signature: the Fourier solution should occupy a flatter, more degenerate loss minimum. Its central quantity, the RLCT λ, measures how degenerate a minimum is. The sparse Fourier solution, with far fewer effective parameters, should be more singular, and weight decay amplifies this preference (Varma et al., 2023; Hoogland et al., 2024). Crucially, degeneracy is multi-faceted: *local curvature* ($\mathrm{tr}(H)$), *effective dimensionality* ($\hat{\lambda}_H$), and *global multiplicity* (SGLD-LLC) probe distinct aspects of the same transition and tell very different stories at grokking.

Contributions.

1. $\mathrm{tr}(H)$ **collapses** $\sim 4\times$ at grokking while **SGLD-LLC is entirely unchanged**, establishing that grokking is a *local*-curvature event invisible to the global multiplicity probe (§4).

2. $\hat{\lambda}_H$ drops only modestly from ~ 24 to ~ 20 post-grokking. This settled value fingerprints the Fourier circuit's intrinsic dimensionality—the **Circuit Imprint** (§4).

3. A Fourier-amplitude analysis of the token embeddings independently finds ~ 11 active frequencies at the same step, agreeing with $\hat{\lambda}_H \approx 20$ despite sharing no information (§4).

2. Background

Local Learning Coefficient. The RLCT λ appears in the free energy as $nF_n = nL_n + \lambda \log n + O(\log\log n)$ (Watanabe, 2013). A lower λ means a more degenerate minimum with a larger effective basin. The LLC $\hat{\lambda}$ estimates λ by running a localised SGLD chain near a trained weight w^*:

$$\hat{\lambda} \;=\; \beta n\, \mathbb{E}_{\mathrm{SGLD}}\big[L(w) - L(w^*)\big], \quad \beta = 1/\log n, \quad (1)$$

with a localisation spring $\frac{\gamma}{2}\|w - w^*\|^2$. Calibration determines which aspect of the RLCT the chain probes: a small step size stays within the local basin and measures local flatness; a large step size drifts between equivalent minima and measures global multiplicity.

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Hessian geometry. We track three scalar quantities derived from the loss Hessian $H = \nabla^2 L(w^*)$:

$$\mathrm{tr}(H) = \sum_i \lambda_i, \quad \lambda_{\max} = \max_i \lambda_i, \quad \hat{\lambda}_H = \frac{\mathrm{tr}(H)}{2\,\lambda_{\max}}. \quad (2)$$

$\mathrm{tr}(H)$ is the total curvature. $\hat{\lambda}_H$ normalises total by peak curvature and serves as an approximation to the number of active constrained dimensions in the loss landscape. We estimate $\mathrm{tr}(H)$ via the Hutchinson finite-difference estimator (Hutchinson, 1990) and $\lambda_{\max}$ via power iteration, both using only forward passes.

3. Setup

We train a one-layer transformer ($d_{\text{model}} = 128$, 4 heads, $\sim$224k parameters) on $(a+b) \bmod 97$ (Power et al., 2022) with a 30% training split, AdamW ($\eta = 10^{-3}$, weight decay 1.0) (Kingma & Ba, 2015; Loshchilov & Hutter, 2019) for 5 000 steps. High weight decay is essential: it penalises the high-norm lookup table faster than the compact Fourier circuit (Varma et al., 2023). All results are mean $\pm$ std over 5 seeds (0–4); grokking completes between steps 2 250 and 3 000 (mean ≈2 700).

At every 250 steps we compute: **(i)** SGLD-LLC via Equation (1) (200 steps + 50 burn-in, $\varepsilon = 3\times10^{-5}$, $\gamma = 100$); **(ii)** $\mathrm{tr}(H)$ and $\hat{\lambda}_H$ via Equation (2) (20 Hutchinson probes, 20 power-iteration steps); and **(iii)** the token-embedding Fourier amplitude spectrum, $A_k = \left(\sum_d |\hat{W}_{k,d}|^2\right)^{1/2}/p$, where $\hat{W}$ is the DFT of W_E along the token axis.

4. Results

Three phases of grokking (Figure 1). *Memorisation:* training loss falls near zero while test loss rises above the random-chance baseline ($\ln 97 \approx 4.6$ nats), confirming the lookup-table solution actively suppresses generalisation. *Plateau:* both losses stagnate. *Grokking:* test loss drops sharply as the Fourier algorithm replaces the lookup table, while training loss stays low. This transition is invisible in training loss alone.

$\mathrm{tr}(H)$ collapses at grokking; SGLD-LLC does not (Figure 2). Figure 2 shows raw absolute values on dual log axes. After an initial rise, SGLD-LLC stabilises at $\sim$1,900 by step $\sim$1,000 and *does not change at the grokking transition*—the global-multiplicity probe is entirely blind to the algorithmic switch. In sharp contrast, $\mathrm{tr}(H)$ **collapses** $\sim$4.1× as the Fourier algorithm replaces the lookup table ($\sim$ 52,000 during the plateau to $\sim$ 12,500 post-grokking). SGLD-LLC is insensitive because global multiplicity—the count of equivalent solutions under DFT symmetries—is similar before and after grokking; only the local basin curvature changes.

The Circuit Imprint: $\hat{\lambda}_H$ fingerprints circuit dimensionality (Figure 3). While $\mathrm{tr}(H)$ collapses $\sim 4\times$, the ratio $\hat{\lambda}_H = \mathrm{tr}(H)/2\lambda_{\max}$ drops only modestly—from $\sim$24 during the plateau to $\sim 20 \pm 2$ post-grokking—because both numerator and denominator shrink in tandem. Crucially, this settled value is highly informative: $\hat{\lambda}_H \approx 20 \approx 11$ active frequencies $\times$ 2 free parameters, recovering the Fourier circuit's intrinsic dimensionality from the loss-surface geometry alone. We term this the **Circuit Imprint**.

Cross-validation via Fourier analysis (Figure 3). The Circuit Imprint makes a testable prediction: if $\hat{\lambda}_H \approx 20$ counts active Fourier dimensions, a direct spectral analysis of the token embeddings should find $\sim$11 active frequencies activating at the same step.

The left panel of Figure 3 confirms this: all frequencies are dim through the plateau; at the grokking transition, $\sim$ 11 frequencies brighten sharply while the rest stay near zero. The right panel overlays the active frequency count (orange) with $\hat{\lambda}_H$ (teal)—both transition at the same step and converge to consistent values. These signals share no information (spectral analysis of the embeddings vs. loss-surface curvature), yet converge simultaneously across all seeds, confirming that loss-surface geometry encodes the circuit's parameter count in a recoverable, task-agnostic form.

5. Discussion

Why SGLD-LLC is blind to grokking. The flat SGLD-LLC reflects a genuine property of the solution space, not a measurement failure. Both the lookup-table and Fourier solutions have high global multiplicity: lookup tables are rearranged by permuting attention heads; Fourier solutions come in families related by DFT symmetries (different active frequency subsets, phase rotations). With the step size used here, the SGLD chain drifts between these equivalent solutions and reports a global count that is approximately the same on both sides of the transition. The *local* basin curvature changes dramatically ($\mathrm{tr}(H)$ collapses 4×), but this signal is invisible at the SGLD's step size.

The Circuit Imprint as a task-agnostic probe. Mechanistic interpretability identifies the Fourier circuit through task-specific projections requiring prior knowledge of the task structure (Nanda et al., 2023). $\hat{\lambda}_H$ arrives at the same quantitative answer via a task-agnostic route: the loss surface carries a curvature signature of the active circuit, readable without knowing which frequencies to look for. The two approaches are complementary—mechanistic interpretability identifies *which* components are active; $\hat{\lambda}_H$ quantifies *how many effective parameters* the solution uses.

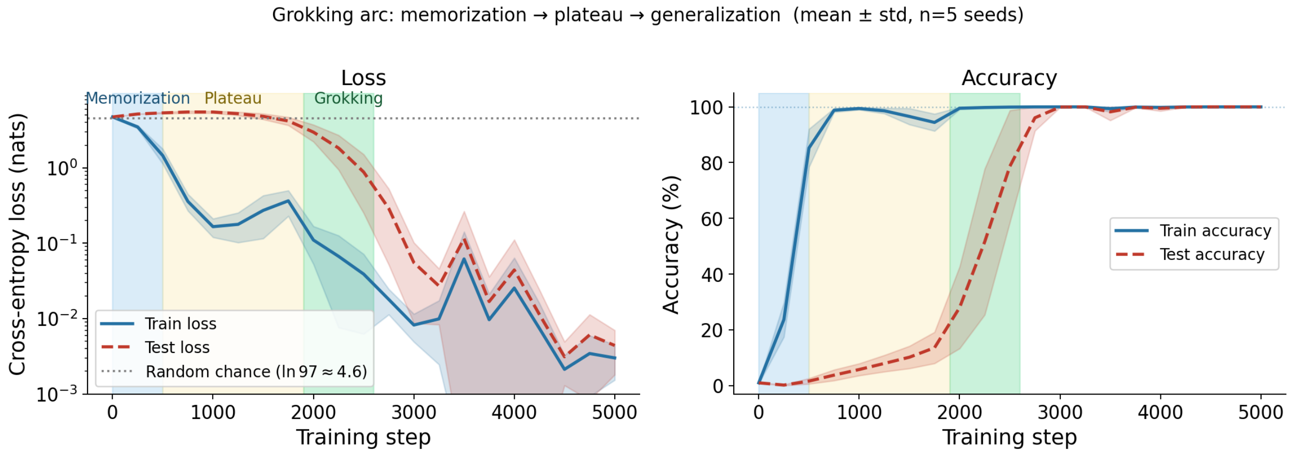

Figure 1. **Three-phase grokking arc** (mean $\pm$ std, 5 seeds). Training accuracy reaches 100% early; test accuracy stays near random chance ($\ln 97 \approx 4.6$ nats) until the abrupt grokking transition (mean step $\approx$2 700). Test loss peaks *above* the random-chance baseline during the plateau, confirming active suppression of generalisation before the algorithmic switch.

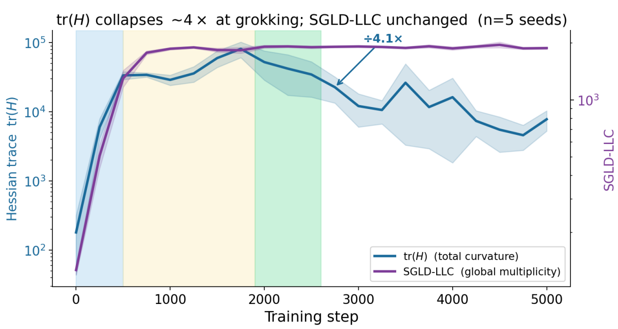

Figure 2. $\mathrm{tr}(H)$ **collapses at grokking; SGLD-LLC does not** (mean $\pm$ std, 5 seeds; both axes log scale). *Left (blue):* $\mathrm{tr}(H)$ rises during memorisation then collapses $\sim$4.1$\times$—from $\sim$52,000 (plateau) to $\sim$ 12,500 (post-grokking). *Right (purple):* SGLD-LLC stabilises at $\sim$ 1,900 and remains flat across the grokking transition. The two probes diverge because $\mathrm{tr}(H)$ captures local basin curvature (which changes) while SGLD-LLC captures global multiplicity (which does not).

Recent work (Author(s) of arXiv:2603.01192, 2026) tracks the LLC as a grokking detector; our results show that $\mathrm{tr}(H)$ is a sharper detector (4$\times$ drop at the transition) while SGLD-LLC is preserved, providing complementary information about basin sharpness vs. solution symmetry.

6. Conclusion

Grokking exposes three distinct geometric signatures that tell surprisingly different stories. $\mathrm{tr}(H)$ collapses $\sim$4$\times$—a clean, threshold-free detector of the algorithmic switch that requires no test-set labels. $\hat{\lambda}_H$ drops modestly from $\sim$ 24 to $\sim$ 20 and encodes the Fourier circuit's intrinsic dimensionality as a geometric fingerprint—the Circuit Imprint. SGLD-LLC is entirely unchanged, measuring global multiplicity which is similar for both solutions. Together, these three probes characterise the transition at three scales: local sharpness, effective dimensionality, and solution symmetry—and demonstrate that loss-surface geometry can serve as a task-agnostic proxy for circuit identification.

Impact Statement

This paper advances the theoretical understanding of generalisation in transformer models. There are no ethical concerns or negative societal consequences associated with this work.

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[To be added in camera-ready version.]

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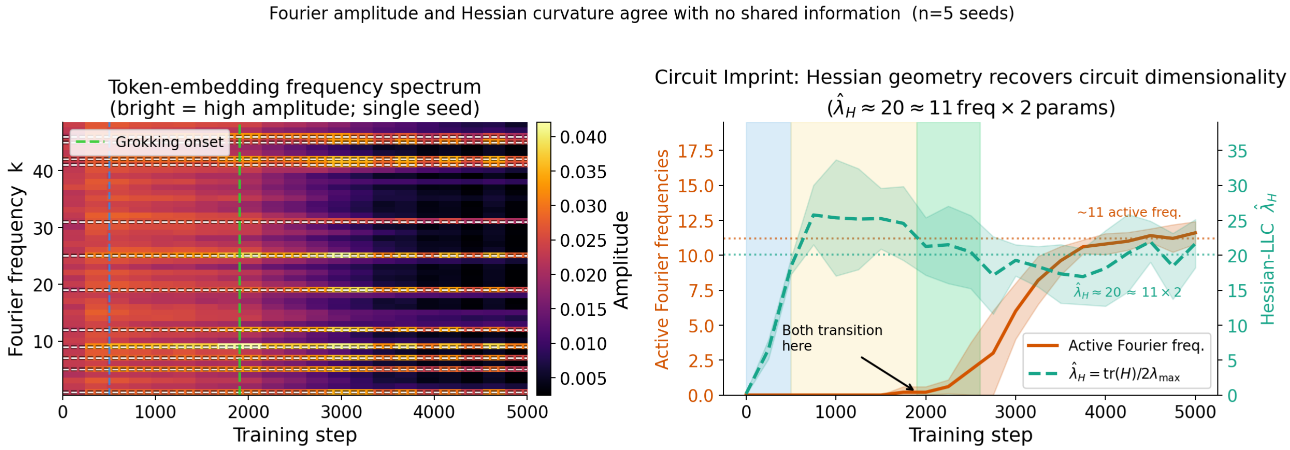

Figure 3. **Circuit Imprint: loss-geometry independently recovers circuit dimensionality** (mean $\pm$ std, 5 seeds). *Left:* Token-embedding Fourier spectrum (single representative seed; bright = high amplitude). At the grokking transition (green dashed line) ~ 11 frequencies activate sharply while all others remain near zero. *Right:* Active frequency count (orange, ~ 11 post-grokking) and $\hat{\lambda}_H$ (teal, 20 ± 2 post-grokking) transition at the same training step and converge to consistent values. The two signals are derived from entirely different objects—embedding spectrum vs. loss-surface curvature—and share no information, yet agree quantitatively across all seeds.

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